

SHOURAV KUMAR MANDAL

21 Lenox Avenue, Paterson, NJ 07502 | 929-301-9368 | shourav.mandal@gmail.com

[Linkedin](#) | [Portfolio](#)

Software Engineer | AI & LLM Systems

Professional Summary:

AI-focused Software Engineer with 5+ years of experience building scalable backend systems and productionizing intelligent applications. Experienced in integrating machine learning and Large Language Model (LLM) services into enterprise environments, including Retrieval-Augmented Generation (RAG) patterns and vector-based retrieval concepts. Strong background in Java (JDK 21), Spring Boot microservices, and RESTful API design, with hands-on experience deploying AI-enabled systems across AWS and GCP using CI/CD automation. Skilled in designing reliable, high-performance architectures that bridge traditional software engineering with modern AI system design within Agile development environments.

Technical Skills:

Programming Languages: Java (JDK 21), Python, SQL, TypeScript

AI & LLM Technologies: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Databases, Embeddings, Prompt Engineering, AI Model Integration, Model Deployment

Backend & Architecture: Spring Boot, Spring Framework, Hibernate, JPA, RESTful APIs, Microservices Architecture, Distributed Systems

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, CloudWatch), GCP, Docker, Kubernetes, Jenkins, CI/CD, Git

Databases: MySQL, Oracle, MongoDB, Redis

Development Practices: Agile, Scrum, Test-Driven Development (TDD), API Design, Performance Optimization

Professional Experience:

Software Engineer - AI & LLM Systems

Oct 2025 – Present

Meta (Facebook) | Menlo Park, California

Project: Contributed to the development and enhancement of the UIGen platform, ensuring seamless front-end and back-end functionality for the deployment and administration of AI-enabled services. Enabled internal teams to integrate, deploy, and monitor machine learning models in production environments. Developed and maintained RESTful APIs and Java-based backend services deployed on AWS to support scalability, data processing, and model inference. Focused on improving system reliability, performance, and data management through CI/CD automation, monitoring solutions, and comprehensive documentation. Worked within Agile environments to align platform capabilities with operational and business requirements.

Responsibilities:

- Transformed machine-learning models created by data science teams into UIGen platform services that are ready for production.
- Designed and implemented Java-based backend services and RESTful APIs to support AI-driven features and administrative workflows.
- Built and managed data ingestion and processing pipelines required for model execution and inference.
- Deployed and maintained scalable AI systems on AWS, guaranteeing dependability under load and real-world usage.
- Troubleshoot production problems, facilitated iterative model upgrades, and tracked model and service performance.
- Created CI/CD pipelines for AI-enabled Java apps to automate the build, test, and deployment procedures.
- Enhanced platform performance and inference reliability by utilizing contemporary Java design patterns and AI integration techniques.
- Improved planning and release cycles with the use of Agile and iterative delivery methods in development workflows.
- For continued platform maintenance, system architecture, service flows, and operational processes were documented.
- To facilitate cross-system communication and AI service consumption, microservices and REST APIs were integrated.
- Using Postman, backend services and AI endpoints were validated to guarantee API dependability and contract compliance.
- Worked together with engineers, data scientists, and product owners to match UIGen capabilities to operational and business objectives.
- Worked directly with cloud-based AI workflows on GCP, incorporating LLM-powered services into pipelines for experimentation and development.

- In order to produce iterative platform enhancements, I actively participated in Agile rituals such as sprint planning, stand-ups, reviews, and retrospectives.

Technologies: Java, Python, Spring Boot, REST APIs, Microservices, AWS, GCP, CI/CD, Jenkins, GitHub, SQL, AI Model Integration

Software Engineer (Computer Systems Analyst/Programmer)

Jun 2022 – Oct 2025

United States Postal Service | Wilkes-Barre, Pennsylvania

Project: Development and upgrading of the USPS Admin Portal user interfaces and the SuperAdmin Functional Interface guarantees perfect front-end and back-end functionality. Internal USPS users can optimize operations through improved postal label design and problem-solving techniques. Created and launched the SuperAdmin Interface, which improves administrative controls and user management by combining front-end and back-end components. primarily focused on enhancing system performance, data management, and user interface.

Responsibilities:

- Developed and oversaw Java backend programs that were scalable, fast, secure, and available to meet changing business needs.
- Deployed a Spring Boot microservice on AWS EC2, implementing RESTful APIs for environment-specific data retrieval and system integration.
- By working with business analysts and cross-functional IT teams, agile approaches for iterative development make it possible to turn challenging software requirements into workable designs.
- In conjunction with industry-standard coding practices, object-oriented programming made sure that systems were maintainable, efficient, and expandable.
- Developed dynamic front-end components using TypeScript, HTML5, CSS3, and Bootstrap, resulting in a responsive UI/UX design and seamless backend service interaction.
- To preserve data integrity and increase query efficiency across platforms, JPA was used to handle connections for database interactions in Spring Boot apps utilizing relational databases (Oracle) and NoSQL (Redis).
- Run several images in Docker to enable containerizing the project in AWS.
- To ensure software dependability, debug and test units.
- Charged with CI/CD pipelines for automatic deployment in addition to monitoring apps hosted on AWS (S3, EC2).
- Working together with IT professionals on software integrations, with a focus on microservices architecture and RESTful APIs to provide seamless data transfer between platforms and apps,
- Ideal for Agile development environments; flexible enough to adjust solutions to meet business objectives through constant stakeholder contact and changing priorities.

Technologies: Java, Typescript, HTML5, React.js, Next.js, MySQL, Jenkins, Visual Studio, Swagger, Github, AWS, Visual Studio

Software Engineer Intern

Jun 2022 – July 2023

SS&C Technology | Manhattan, NYC

Project: Created a web-based solution to enhance the management and querying skills of healthcare professionals. A smooth experience depends on the frontend and backend of Java interacting better. streamlined the site's design to improve user experience and interface, speed up SQL queries, and retrieve data in response to user input. Increased data validation to prevent SQL injection and other security vulnerabilities.

Responsibilities:

- Completed research spikes to look into new technologies and solutions while working on assigned user stories as part of Agile sprints.
- In order to ensure a seamless connection between the user interface and Java-based backend services, frontend and backend components were created using RESTful APIs and microservices architecture.
- Enhanced SQL queries and established database interactions to boost data retrieval performance, ensuring rapid and efficient querying with minimal delay.
- Executed data validation and sanitization methods, leading to a decrease in security threats such as SQL injection attacks, while bolstering the overall security of the platform and the integrity of its data
- Redesigned the homepage and incorporated UI/UX best practices, improving user engagement among medical professionals through a responsive and intuitive interface.
- Used GitHub for source code management, which promoted effective branching techniques, permitted collaborative development, and guaranteed version control.
- Before deploying, extensive unit tests and debugging were conducted using Jira and Eclipse to ensure the code was reliable and high-quality.

Technologies: MySQL, Spring Framework, Legacy, Java Development, JavaScript, Jira, Eclipse, AWS, and GitHub

System Production Support Intern

Jan 2022 - Jun 2022

NYC Department of Buildings (DOB) | Broadway, NYC

Project: Developed a solution for tracking data management and system performance metrics. Excel, SQL, and Python were used to process, visualize, and report data. Power BI and Power Automate enhanced productivity by automating processes.

Responsibilities:

- Collaborated to understand and release new features and data requirements with developers and business analysts.
- Connected with clients of both technical and non-technical backgrounds, eliciting details of problems and providing clear solutions.
- Used libraries like Pandas, NumPy, and Matplotlib for complex data processing and visualization, and Python for data automation.
- Developed and optimized SQL queries for data extraction, ensuring efficient insight extraction from large datasets and improving report generation.
- Committed to ensuring the accuracy and dependability of datasets for reporting and decision-making through data integrity, validation, and cleanliness.

Technologies: Excel, Python (Pandas, NumPy, Matplotlib), SQL, Microsoft Power BI, Power Automate

Support Tech Team Member

Aug 2021 - Jan 2022

Herbert H Lehman College (CUNY) | Bronx, NYC

Responsibilities:

- Helped faculty incorporate HyFlex learning to ensure smooth interaction between in-person and remote students.
- Provided classroom technology technical support, ensuring that microphones, cameras, and audio/video systems operated without a hitch before and after courses.
- To ensure that class activities were disrupted as little as possible, I helped teachers diagnose and fix hardware and software issues.
- Helped educators identify and resolve hardware and software issues, ensuring minimal disruption to classroom activities.
- Worked with the IT department to guarantee that the most recent software updates and security patches were implemented on classroom technology and learning platforms

Technologies: Windows, macOS, Zoom, Microsoft Teams, Classroom AV Systems, Blackboard LMS

Education:

Colorado State University, Fort Collins

Sep 2024 - Jun 2026

Master of Science in Computer Science (AI & Machine Learning)

CUNY Lehman College, New York

Aug 2019 - May 2024

Bachelor of Science in Computer Science

Bronx Collaborative High School, New York

Aug 2016 – Jun 2019

High School Diploma